

COMPONENTS

Design kit for USB charger

Power Integrations (PI) has announced RDK-420, a new reference design kit for a 10W CV/CC USB charger based on the company's InnoSwitch-CH family of highly-integrated switcher ICs, which combine the primary-side switch together with primary and secondary controllers and feedback circuits into a single surface-mount package.

The ICs leverage accurate secondary-side regulation (SSR) using PI's high-speed digital FluxLink technology to communicate direct voltage and current measurements across the safety isolation barrier. This technique permits precise control without the need of a bulky optocoupler, while avoiding performance compromises inherent in primary-side regulation (PSR).

Power Integrations

Website: www.power.com

Satellite demodulator chip

STMicroelectronics has announced the world's first 500Mbaud high-symbol-rate (HSR) satellite demodulator chip. The STiD135 enables efficient bandwidth usage and increased throughput in satellite Internet provisioning when deployed with transponders aimed at using the higher-frequency bands to send data via Ka-band communication satellites.

STMicroelectronics

Website: www.st.com

Digital-output switch

Maxim Integrated addresses the need for higher throughput, smaller footprint, and lower power in industrial automation with MAX14900E digital-output switch.

This 8-channel IC provides up to 70x faster speed, uses over 20 per cent

less power and occupies 40 per cent less space than existing solutions. It is controlled by an 8-bit parallel interface or via a standard serial peripheral interface (SPI), which is also used for configuration and monitoring.

Maxim Integrated

Website: www.maximintegrated.com

3D magnetic sensor

Infinion's 3D magnetic sensor TLV493D-A1B6 features highly-accurate 3D-sensing with extremely low power consumption. It suits consumer and industrial applications such as joysticks, control elements used for white goods and multi-function knobs, and electric meters where it helps to protect against tampering.

Infinion Technologies

Website: www.infineon.com

Connector

The 3mm pitch RotaConnect WtB connector from FCI mates horizontally with the existing board-to-board version, which is hermaphroditic and surface mount soldered to the PCB. It also features a dual-beam-spring contact design, delivering high-performance and reliability.

RotaConnect WtB supplies power to an LED PCB in a string of rigid PCB light strips. The connector is ideal for control boards, sensors, actuators, rigid LED strips and automatic lighting.

FCI Electronics

Website: www.fci.com

Memory kit

HyperX, a division of Kingston Technology, has created the world's fastest DDR4 128GB memory kit that runs at 3000MHz.

The kit consists of eight 16GB HyperX Predator modules (16GB x 8) with ultra-tight 16-16-16-36 timings and XMP profiles for easy and stable overclocking. This is achieved

using MSI X99 MPOWER motherboard in an 8-module, quad-channel configuration along with an Intel Core i7 5820K processor.

Kingston Technology

Website: www.kingston.com/en/hyperx

Capacitor

Vishay Intertechnology has introduced a new thin-film bar MOS capacitor designed for use in high-power hybrid assemblies and SiC and GaN applications. The Vishay Dale resistors electro-films BRCP offers high power handling and operating voltage up to 100V in two compact case sizes, measuring 120mil by 35mil (case A) and 240mil by 35mil (case B), allowing for smaller product designs without sacrificing performance.

With its high working voltage over a capacitance range of 5pF to 100pF and its robust MOS construction, the device is optimised for LC, RC or LRC filters, RF choking and DC blocking and impedance matching. For these applications, the BRCP can accommodate multiple wire bonds, with seven bonds for case A and 15 bonds for case B, at their lowest values.

Vishay Intertechnology Inc.

Website: www.vishay.com

EtherCAT slave controller

Microchip Technology has announced the standalone EtherCAT slave controller with two 10/100 PHYs, the LAN9252. This highly-integrated device's dual 10/100 Ethernet transceivers support both fibre and copper, along with cable diagnostics capabilities. The LAN9252

supports traditional host bus and SPI/SQI communication, along with standalone digital input/output (I/O) interfaces, providing system designers the flexibility to select from a wide range of microcontrollers when implementing the real-time EtherCAT communications standard.

Microchip Technology Inc.

Website: www.microchip.com

